Analytical Cume
April 6, 2012

The main thing we do in Analytical chemistry is measure. If you allow two chemicals to react, say you place metallic zinc in a solution of copper sulfate, copper will plate out on the zinc. The reaction being:

$$Cu^{2+} + Zn \leftrightarrow Cu + Zn^{2+}$$

Maybe you could do some kinetics with this by looking at the changes in concentrations over time, but there is really no way to measure the changes in energy.

The problem has been traditionally solved by separating reactant/product pairs, placing them in separate containers called cells. The line notation for the cell given by:

$Zn|ZnSO_4$ (1.0 M) || $CuSO_4$ (1.0 M) |Cu

The potential difference between the zinc and copper electrodes is measured by a meter with a large internal resistance, red lead placed on the copper electrode, black to the Zinc.

1. With the meter connected, will the concentrations in the cell change much over about 1 hour? (yes or no answer will do)
2. If the meter reads +1.1 V write the overall red-ox reaction for the cell.
3. Is this system at equilibrium?

The meter is removed and a wire with almost no resistance is connected between the copper and zinc electrode.

4. With the wire connected, will the concentrations in the cell change much over about 1 hour? (yes or no answer will do)
5. After 1 hour, the wire is removed and the meter reconnected. The meter now reads 0.0 V. Calculate the concentrations of Zn^{2+} and Cu^{2+} in the beakers. Assume there is plenty of the solid metal.
6. Is this system at equilibrium?
7. What are the differences between the states in #3 and #6.

More Info:

$Cu^{2+} + 2e^- \leftrightarrow Cu$ $\quad E^o = 0.339$ V

$Zn^{2+} + 2e^- \leftrightarrow Zn$ $\quad E^o = -0.762$ V

$$E^o_{rxn} = \frac{0.0591}{n} \log K$$